

SQLite with Dart

File-Based Relational Database

What is SQLite?

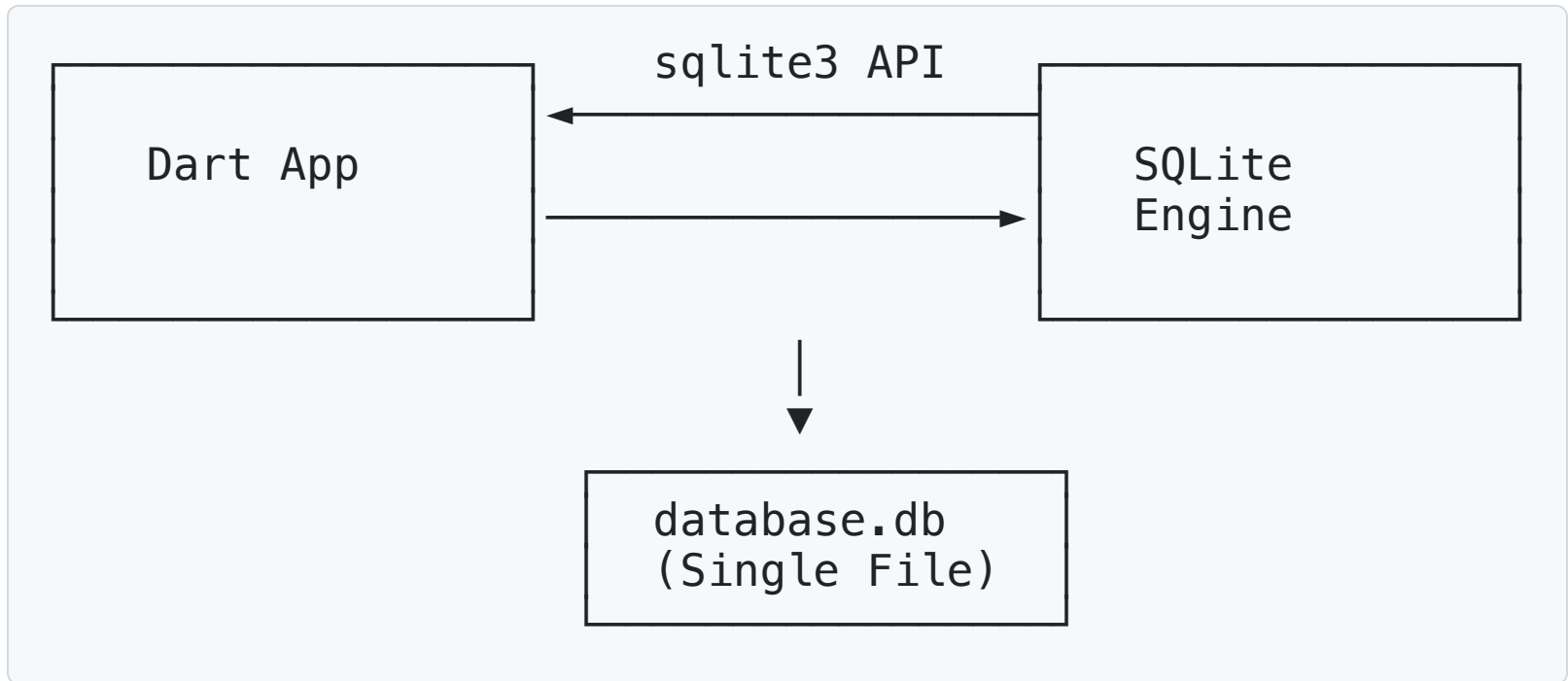
- **File-based database** - Single file contains entire database
- **Relational database** with full SQL support
- **Zero configuration** - No server setup required
- **Embedded** in applications (mobile, desktop, web)
- **Most widely deployed** database engine in the world

Key Characteristics:

- Full ACID compliance (transactions)
- Complete SQL implementation
- Cross-platform compatibility
- Public domain (no licensing fees)
- Self-contained C library

Used by: Pocketbase, Android, iOS, browsers, desktop apps, embedded systems

Architecture Overview



Dart SQLite

- Using SQLite in Dart is making SQLite through the helper functions.
- We cannot use SQLite for client side web applications.
 - We can use SQLite for server side web applications.
 - For client side web application, we can use IndexedDB.

Project Setup

pubspec.yaml

```
dependencies:  
  sqlite3: ^2.4.0      # Pure Dart SQLite interface  
  path: ^1.8.3         # Cross-platform path handling
```

```
import 'package:sqlite3/sqlite3.dart';  
import 'package:path/path.dart' as p;  
  
final dbPath = p.join('data', 'my_database.db');  
final db = sqlite3.open(dbPath);
```

SQLite execute and select

Understand how to run SQL commands safely and efficiently.

- Use "db.execute" to **change** data.
- Use "db.select" to **retrieve** data.
- Use "prepare + execute/select" to stay **safe and efficient**.

- *Quick Execution:* `db.execute`
- ♦ Runs raw SQL directly
- ♦ Good for one-time commands
- ⚠ Risk: SQL injection if inputs are embedded directly

```
db.execute("INSERT INTO students (name, age)  
          VALUES ('Alice', 22)");
```


- Safe & Reusable: `db.prepare + stmt.execute`
-  Uses placeholders (`?`) to prevent injection
-  Efficient for repeated queries
-  Requires cleanup with `dispose()`

```
final stmt = db.prepare(  
    "INSERT INTO students (name, age) VALUES (?, ?)");  
stmt.execute(['Alice', 22]);  
stmt.dispose();
```

db.select vs db.execute

Feature	db.execute	db.select
Purpose	INSERT/UPDATE/DELETE	SELECT queries
Returns	Nothing or metadata	List of rows/maps
Example	<code>db.execute("DELETE ...")</code>	<code>db.select("SELECT * FROM ...")</code>

- For Queries, use `db.select` .
- For other actions, use `db.execute` .

FooBar Data Model

- The FooBar class

```
import 'package:sqlite3/sqlite3.dart';  
class FooBar {  
  String foo;  
  int bar;  
  FooBar({  
    required this.foo,  
    required this.bar,  
  });
```

Conversion from/to SQLite

```
factory FooBar.fromRow(Row row) {  
    return FooBar(  
        foo: row['foo'] as String,  
        bar: row['bar'] as int,  
    );  
}  
  
Map<String, dynamic> toMap() => {  
    'foo': foo,  
    'bar': bar,  
};
```

The == operator for object comparison

```
@override
bool operator ==(Object other) {
    if (identical(this, other)) return true;
    return other is FooBar && other.foo == foo && other.bar == bar;
}

@override
int get hashCode => foo.hashCode ^ bar.hashCode;
}
```

SQLite Dart Programming

FooBarCrudSQLite is the class for SQLite programming.

```
Database? _database;  
final String _databaseName = 'foobar.db';  
final String _tableName = 'foobars';  
final String _dataDirectory = 'data';
```

database property

```
Future<Database> get database async {  
    if (_database != null) return _database!;  
  
    // Ensure data directory exists  
    final dataDir = Directory(_dataDirectory);  
    if (!await dataDir.exists()) {  
        await dataDir.create(recursive: true);  
    }  
  
    String dbPath = join(_dataDirectory, _databaseName);  
    _database = sqlite3.open(dbPath);  
    _createTableIfNotExists();  
    return _database!;  
}
```

create table

- Fromm the database, we create table if not exists.

```
void _createTableIfNotExists() {  
    final db = _database!;  
    db.execute(''  
        CREATE TABLE IF NOT EXISTS $_tableName (  
            id INTEGER PRIMARY KEY AUTOINCREMENT,  
            foo TEXT NOT NULL,  
            bar INTEGER NOT NULL,  
            created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
        )  
        '' );  
}
```


CRUD

- CREATE: INSERT INTO
- READ: SELECT * FROM
- UPDATE: UPDATE
- DELETE: DELETE FROM

CREATE

```
Future<int> create(FooBar foobar) async {  
    final db = await database;  
  
    final stmt = db.prepare('''  
        INSERT INTO $_tableName (foo, bar)  
        VALUES (?, ?)  
    ''');  
  
    stmt.execute([foobar.foo, foobar.bar]);  
    stmt.dispose();  
  
    return db.lastInsertRowId;  
}
```

Read

```
/// READ: Get a FooBar by ID
/// Returns null if not found
Future<FooBar?> read(int id) async {
    final db = await database;

    final stmt = db.prepare(
        'SELECT * FROM $_tableName WHERE id = ?');
    final result = stmt.select([id]);
    stmt.dispose();

    if (result.isEmpty) return null;

    return FooBar.fromRow(result.first);
}
```

read all

```
/// READ: Get all FooBar records
/// Returns a list of all FooBar objects
Future<List<FooBar>> readAll() async {
    final db = await database;

    final ResultSet resultSet =
        db.select('SELECT * FROM $_tableName ORDER BY id');

    return resultSet.map((row) =>
        FooBar.fromRow(row)).toList();
}
```

Find By

```
/// READ: Find FooBar records by foo field (like a search)
/// Returns a list of matching FooBar objects
Future<List<FooBar>> findByFoo(String foo) async {
  final db = await database;

  final stmt = db.prepare(
    'SELECT * FROM $_tableName WHERE foo LIKE ? ORDER BY id');
  final result = stmt.select(['%$foo%']);
  stmt.dispose();

  return result.map((row) => FooBar.fromRow(row)).toList();
}
```

Update

```
/// UPDATE: Update an existing FooBar record
/// Returns true if successful, false if record not found
Future<bool> update(int id, FooBar foobar) async {
    final db = await database;
    final stmt = db.prepare('''
        UPDATE $_tableName
        SET foo = ?, bar = ?
        WHERE id = ?
    ''');
    stmt.execute([foobar.foo, foobar.bar, id]);
    stmt.dispose();
    return db.updatedRows > 0;
}
```

Delete

```
/// DELETE: Remove a FooBar record by ID
/// Returns true if successful, false if record not found
Future<bool> delete(int id) async {
    final db = await database;
    final stmt = db.prepare(
        'DELETE FROM $_tableName WHERE id = ?');
    stmt.execute([id]);
    stmt.dispose();
    return db.updatedRows > 0;
}
```

Delete All

```
/// DELETE: Remove all FooBar records (use with caution!)
/// Returns the number of deleted records
Future<int> deleteAll() async {
    final db = await database;
    db.execute('DELETE FROM $_tableName');
    return db.updatedRows;
}
```


SQL Query Fundamentals

Basic SELECT Queries

```
-- Get all students, ordered by ID
SELECT * FROM students ORDER BY id;

-- Get specific student by ID (parameterized)
SELECT * FROM students WHERE id = ?;

-- Get students by major
SELECT * FROM students WHERE major = 'Computer Science';

-- Get students in age range
SELECT * FROM students WHERE age BETWEEN 18 AND 25;
```

```
-- Count students by major  
SELECT major, COUNT(*) as count  
FROM students  
GROUP BY major;
```

```
-- Get average age  
SELECT AVG(age) as average_age FROM students;
```

Advanced Queries

```
-- Search by partial name match
SELECT * FROM students WHERE name LIKE '%John%';

-- Multiple conditions
SELECT * FROM students
WHERE age > 20 AND major = 'Computer Science';

-- Ordering and limiting results
SELECT * FROM students
ORDER BY age DESC, name ASC
LIMIT 10;
```

foobar project

```
sqlite/
├── lib/
│   ├── models/
│   │   └── foobar.dart                # Data model
│   └── services/
│       └── foobar_crud_sqlite.dart    # CRUD operations
├── test/
│   └── foobar_crud_test.dart          # Unit tests
├── data/
│   └── foobar.db                      # SQLite database
└── doc/
    └── crud_tutorial.md               # Documentation
```

Development Workflow

Running the Code

- Run the Application

```
cd /path/to/sqlite/project  
dart run lib/main.dart
```

- Run the Tests

```
dart test
```

- Run Specific Test

```
dart test test/foobar_crud_test.dart
```

- Run with Verbose Output

```
dart test --reporter=expanded
```

- Check the Database File

```
# Database is created in data/ directory  
ls -la data/  
# Explore with SQLite CLI (if installed)  
sqlite3 data/foobar.db
```

- You can use the SQLite and SQLite Viewer VSCode extension.

Databases

Feature	PocketBase	Firebase	SQLite	IndexedDB
Type	Server + SQLite	Cloud NoSQL	File-based SQL	Browser NoSQL
Location	Self-hosted	Google Cloud	Local file	Browser storage
Real-time	✓ Built-in	✓ Built-in	✗ None	✗ None
Authentication	✓ Built-in	✓ Complete	✗ Manual	✗ Manual
Scalability	⚠ Manual	✓ Automatic	✗ Single user	✗ Single user
Queries	✓ REST API	✓ Rich NoSQL	✓ Full SQL	✗ Key-value
Offline	✗ Network only	✓ Smart sync	✓ Always	✓ Always
Cost	 Free hosting	 Pay-per-use	 Free	 Free

Decision Framework

- **Choose PocketBase** for: Self-hosted real-time apps, educational projects, MVPs, data control`
- **Choose IndexedDB** for: Browser-only applications, offline-first web apps, client-side caching



Choose SQLite for: Single-user apps, offline-first, embedded applications

- **Choose Firebase** for: Global scale, automatic scaling, rapid development without hosting

SQLite Limitations

Concurrency Limitations:

- Single writer at a time
- Read-heavy workloads perform better
- Not ideal for high-concurrency applications
- Limited network database access

Scale Limitations:

- Database size practical limit (~281 TB theoretical)
- Single database file can become large
- No built-in replication or clustering
- Limited user management features

Feature Limitations:

- No stored procedures or user-defined functions
- Limited data types compared to full SQL databases
- No Complex JOIN
 - No RIGHT OUTER JOIN or FULL OUTER JOIN

SQLite as the most famous program

Why SQLite is Special

- **Most widely deployed database in the world**
- **Runs in browsers, phones, IoT devices, apps, operating systems**
- **Used by billions of devices daily**

Success & Support

- **Zero admin:** Works out-of-the-box with no setup
- **Public domain:** Free for everyone, no license worries
- **Trusted by industry:** Apple, Google, Microsoft, Adobe, and more
- **Vibrant community:** Strong public support and documentation